Qualification Exam

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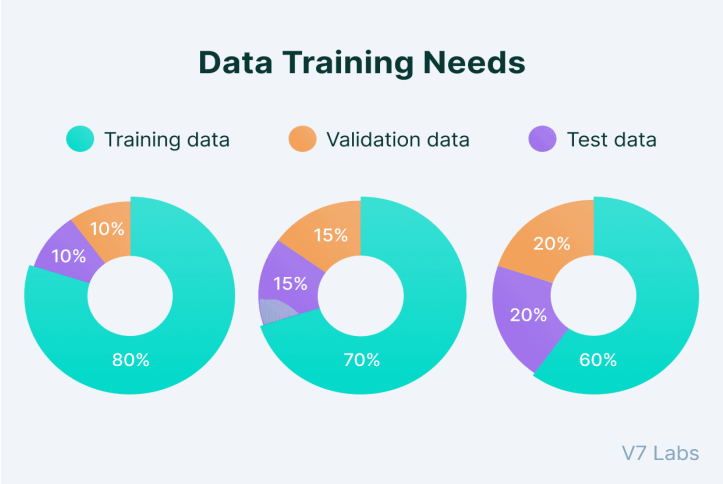
Software Analysis & Intelligence Lab

JLeaks: A Featured Resource Leak Repository Collected From Hundreds of Open Source Java Projects

- Memory leaks are not Resource leaks!
- High quality leaks from real projects
- Code duplication
- Availability
- Garbage Collector the misunderstood program!

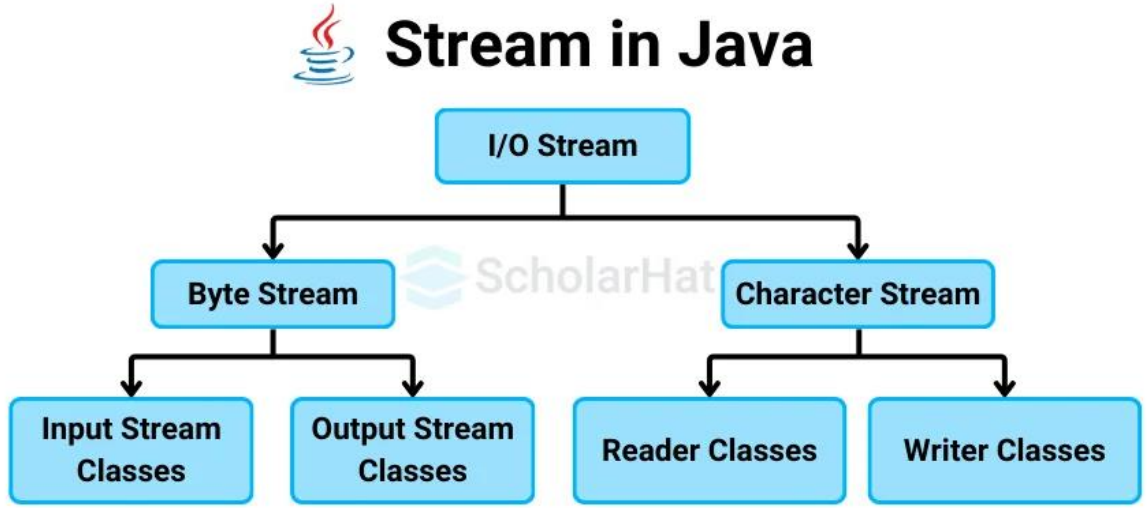






Resource leaks

- File
- Sockets
- Threads



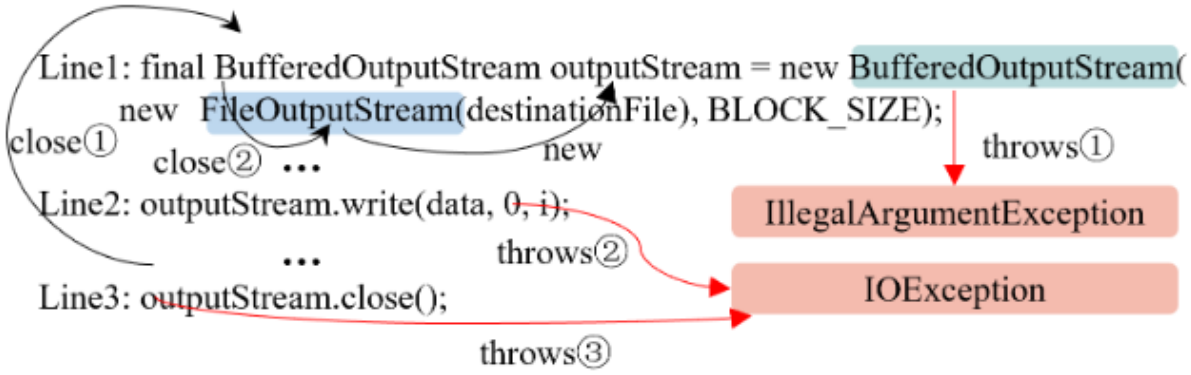


Figure 2: A file resource leak from PhoenicisOrg/phoenicis.

Approach

- Collection
- Annotation
  - 1,094 defects from 321 projects
- Compilation
  - 86 projects with 254 defects

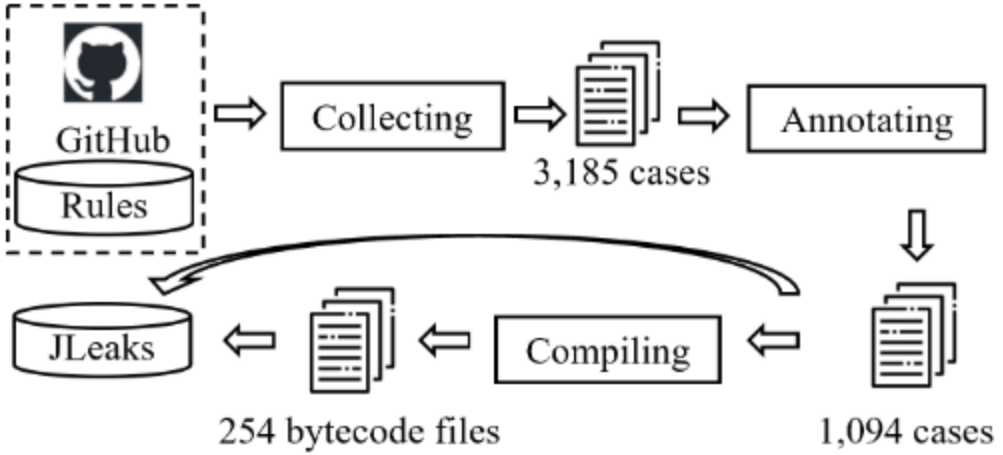


Figure 3: The workflow of building JLeaks.

Table 2: Cohen's Kappa metric.

Dimension	Cohen's Kappa
resource leak	0.88
resource type	0.74
root cause	0.74
fix approach	0.92

Table 1: Rules for filtering projects and commits

Rules for project	Keywords for commits	
	Message	Code
# of commits > 20	leak	close
# of issues > 20	leakage	stop
# of contributors > 10	close	shutdown
# of releases > 0	closed	try(*)
# of stars > 500	release	CloseUtil
duration > 50 weeks	resource	-

Defects Distribution

- 797(73%) file leaks
- 279(26%) socket leaks
- and 18(2%) thread leaks

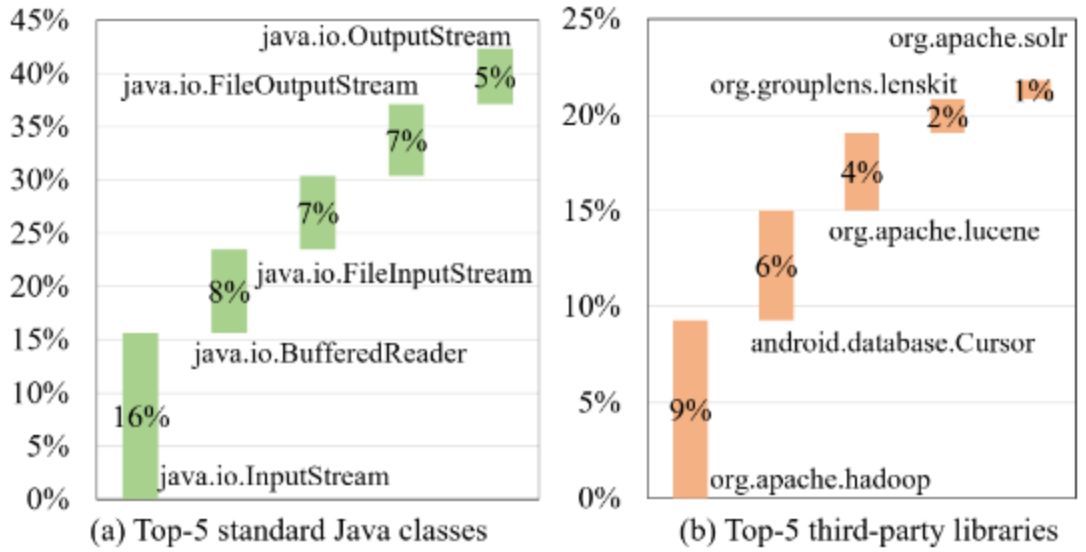
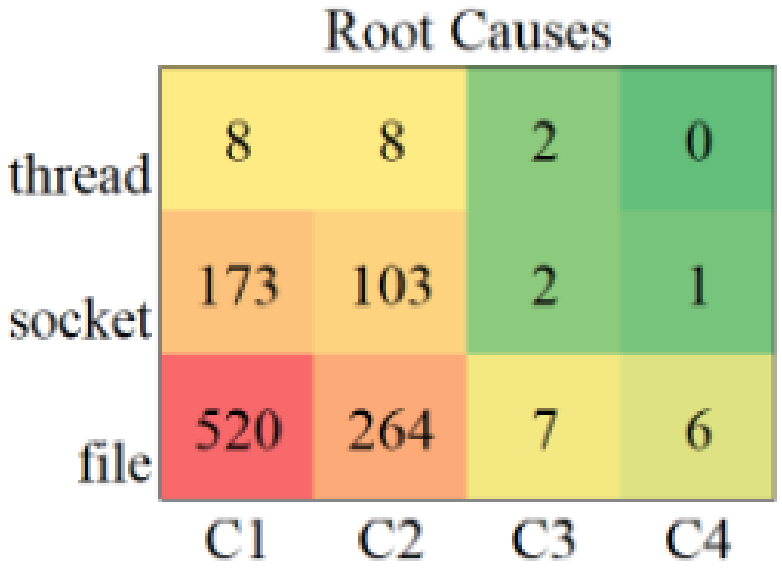


Figure 5: Top ranked libraries of resource leaks in JLeaks.

Root Causes

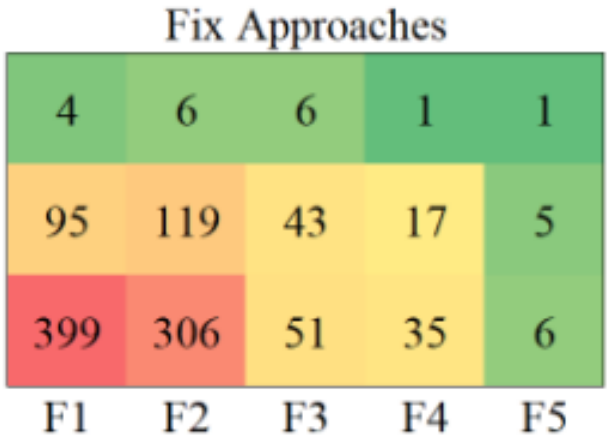
- Intra-procedure,
- Inter-procedure,
- and inter-library
  - NoCloseEPath 64%
  - NoCloseRPath 34%
  - close()
  - noCloseCPath.



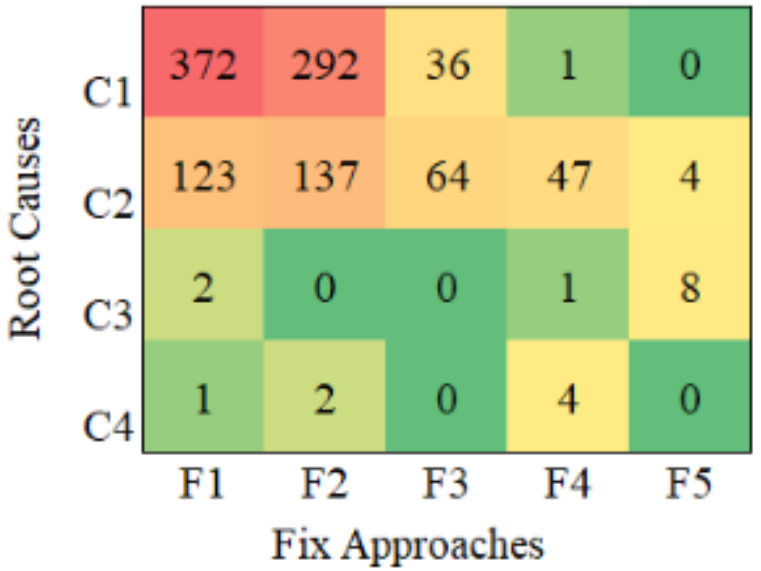
Fix Approaches

Table 4: Fix approaches for resource leak.

Short	Fix Approaches	Description	# defects
F1	try-with	Use try-with-resources	498
F2	CloseInFinally	Close in finally	431
F3	CloseOnEPath	Close on exception paths	100
F4	CloseOnRPath	Close on regular paths	53
F5	AoRClose	Add or rewrite close	12



Patch Correctness



Repository Comparison

Table 5: Informativeness & completeness comparison.

Repository	Project	Type	Link	Location	Root cause	Source files	Bytecode	Patch	Patch correctness
DroidLeaks	✓	×	✓	○	✓	○	✓	○	×
Leak19	✓	✓	✓	×	✓	×	×	○	×
JLeaks	✓	✓	✓	✓	✓	✓	✓	✓	✓

Table 6: Distribution of resource types in three repositories.

Repository	# file	# socket	# thread
DroidLeaks	108	**182**	2
Leak19	129	51	**73**
JLeaks	**797**	**279**	18

Repository Comparison

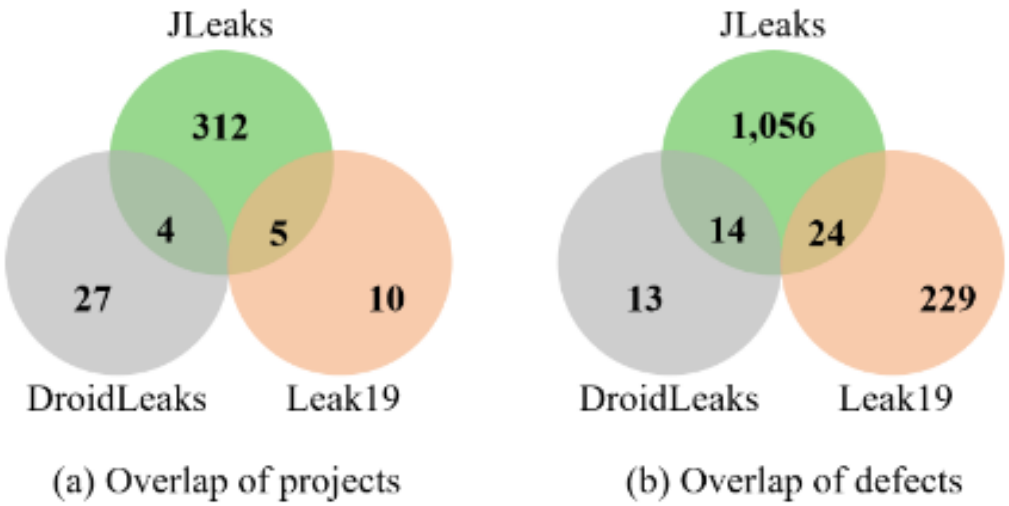


Figure 9: Overlap between JLeaks, DroidLeaks, and Leak19.

Application & Conclusion

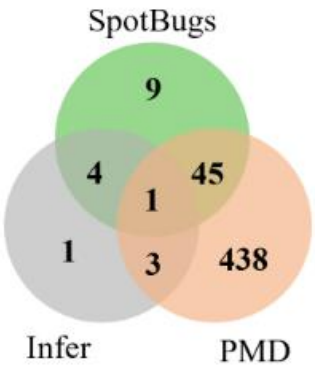


Figure 10: Overlap of defect detection results between PMD, SpotBugs, and Infer.

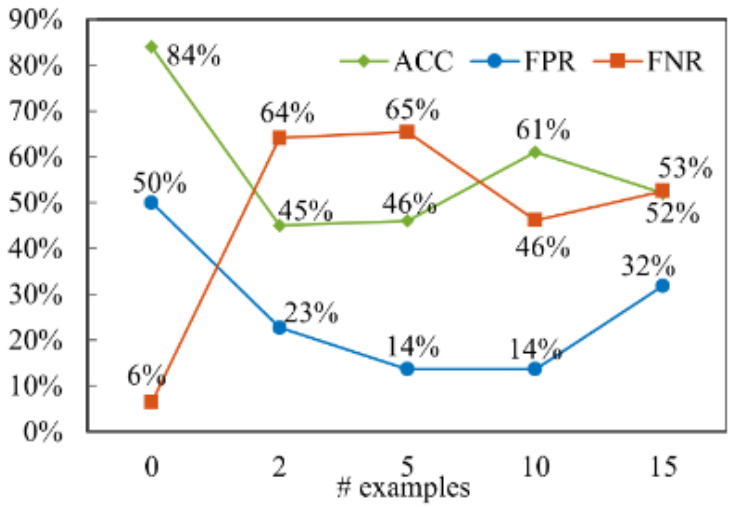


Figure 11: Commits classification results of GPT-3.5 with different numbers of examples in prompt.

Large Language Models for Software Engineering: A Systematic Literature Review

- LMs vs LLMs
- Software Engineering and LLMs
  - Code, data, text analysis
  - Codex with 12 Billion Parameters
  - ChatGPT-4 (OpenAI)

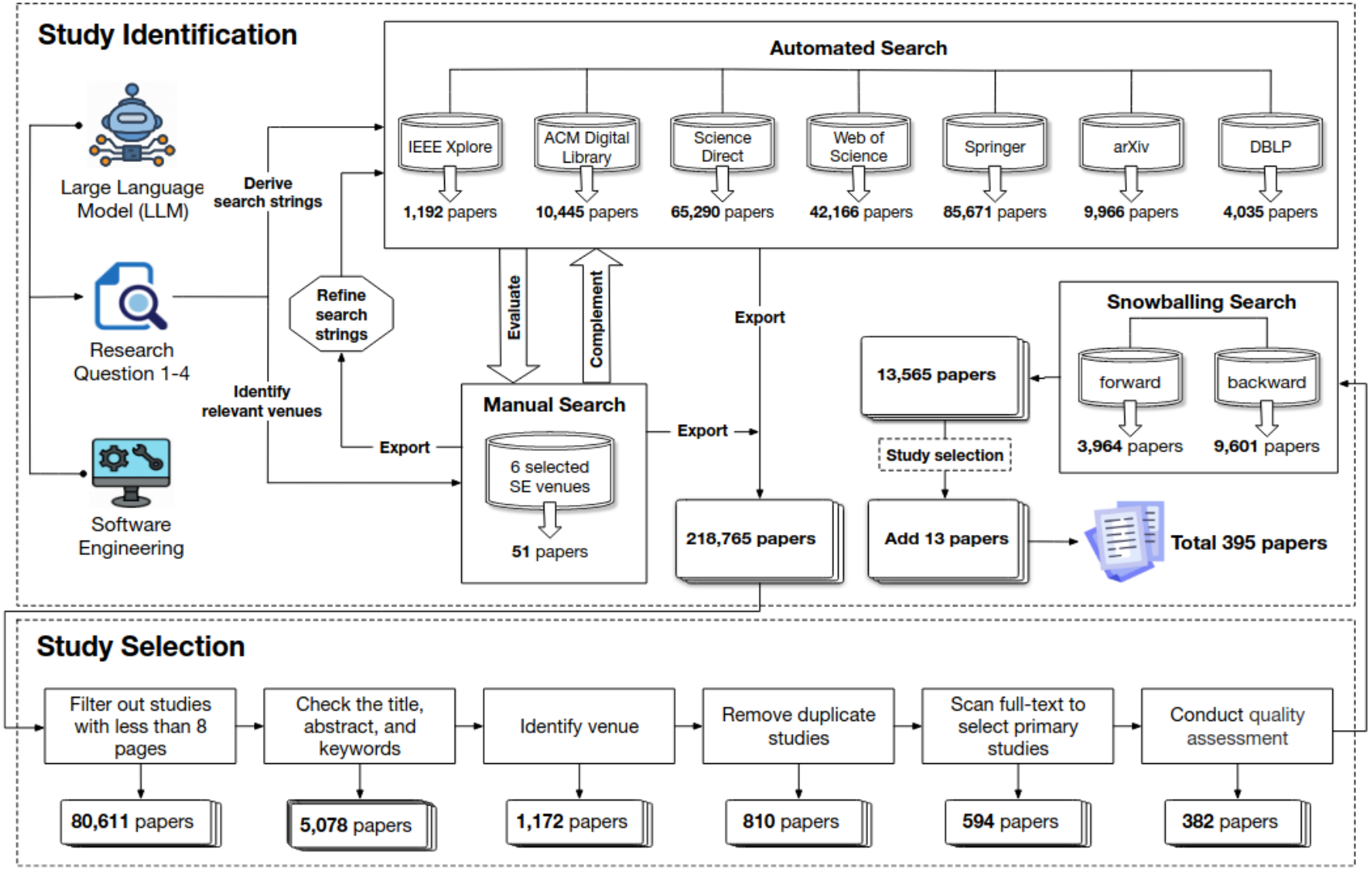
Related Work

Table 1. State-of-the-art surveys related to LLMs for SE.

Reference	Year	Scope of models[1]	Scope of SE tasks	SLR[2]	Time frame	# Collected Papers
Zhang *et al.* [543]	2023	Code LLM	Automated program repair	✓	2017-2023	185
Zheng *et al.* [564]	2023	Code LLM	General SE scope	✓	2021-2023	149
Fan *et al.* [85]	2023	LLM	General SE scope	×	-	Not specified
Zan *et al.* [526]	2023	LLM (12M+)	NL2Code	×	2020-2023	Not specified
Wang *et al.* [448]	2023	LLM (117M+)	Software testing	✓	2019-2023	52
Wang *et al.* [452]	2022	ML, DL[3]	General SE scope	✓	2009-2020	1,209 (ML) + 358 (DL)
Yang *et al.* [509]	2022	DL	General SE scope	✓	2015-2020	250
Watson *et al.* [466]	2022	DL	General SE scope	✓	2009-2019	128
Our work	2024	LLM	General SE scope	✓	2017-2024	395

[1] "M" means million and "B" means billion. The numbers in parentheses indicate the parameter sizes of LLMs.
[2] SLR stands for Systematic Literature Review. This column denotes whether the paper follows an SLR process.
[3] ML and DL refer to Machine Learning and Deep Learning, respectively.

Study identification and selection process



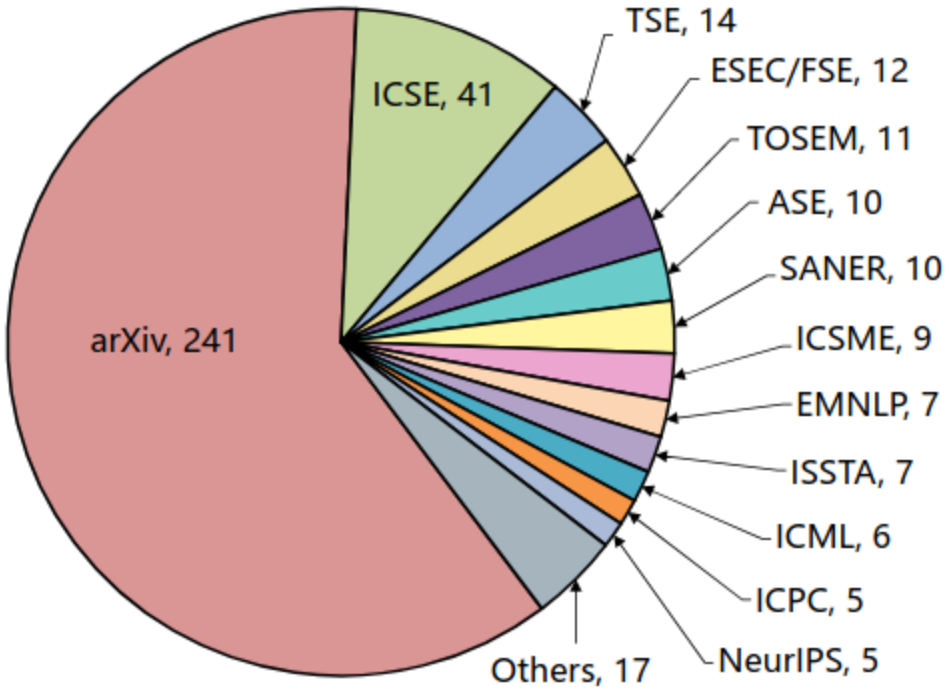
Inclusion criteria and Exclusion criteria

Inclusion criteria
1) The paper claims that an LLM is used.
2) The paper claims that the study involves an SE task.
3) The paper with accessible full text.
Exclusion criteria
1) Short papers whose number of pages is less than 8.
2) Duplicate papers or similar studies with different versions from the same authors.
3) Studies belonging to books, thesis, monographs, keynotes, panels, or venues not executing a full peer-review process.
4) Tool demos and editorials.
5) The paper is published in a workshop or a doctoral symposium.
6) The paper is a grey publication, e.g., a technical report or thesis.
7) Non-English written literature.
8) The paper mentions the use of LLMs without describing the employed techniques.
9) The paper leverages SE methods to enhance LLMs, rather than focusing on using LLMs for SE tasks.

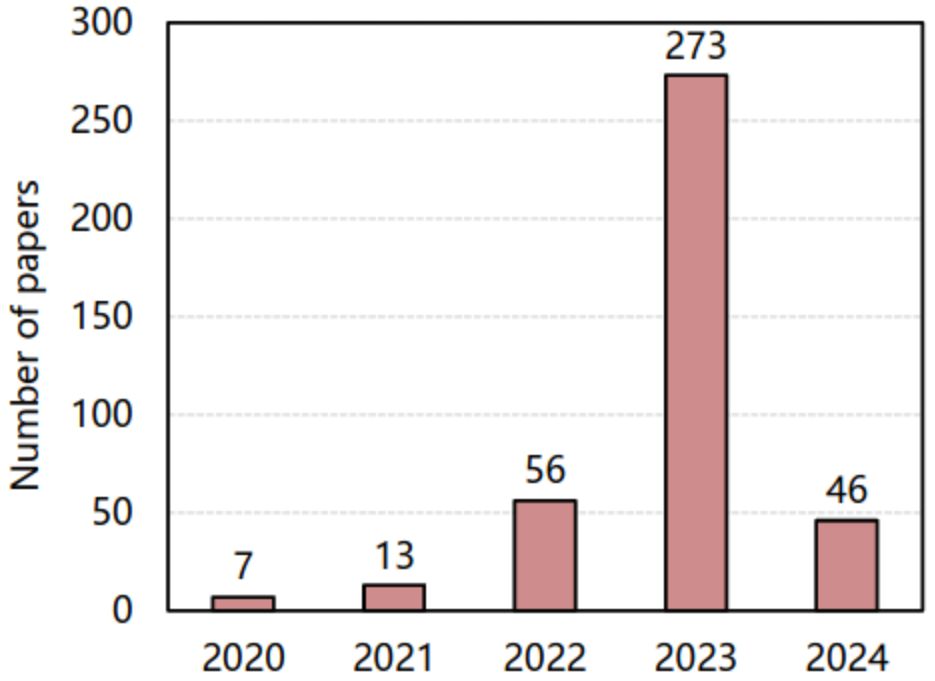
Checklist of Quality Assessment Criteria (QAC) for LLM studies in SE.

ID	Quality Assessment Criteria
QAC1	Is the study relevant to SE tasks?
QAC2	Does the study utilize LLMs?
QAC3	Is the research not a secondary study, such as an SLR, review, or survey?
QAC4	Was the research published in a high-repute venue?
QAC5	Is there a clear motivation for the research?
QAC6	Does the study provide a clear description of the techniques used?
QAC7	Are the experimental setups, including experimental environments and dataset information, described in detail?
QAC8	Does the study clearly confirm the experimental findings?
QAC9	Are the key contributions and limitations of the study discussed?
QAC10	Does the study make a contribution to the academic or industrial community?

Overview of the selected 395 papers' distribution.

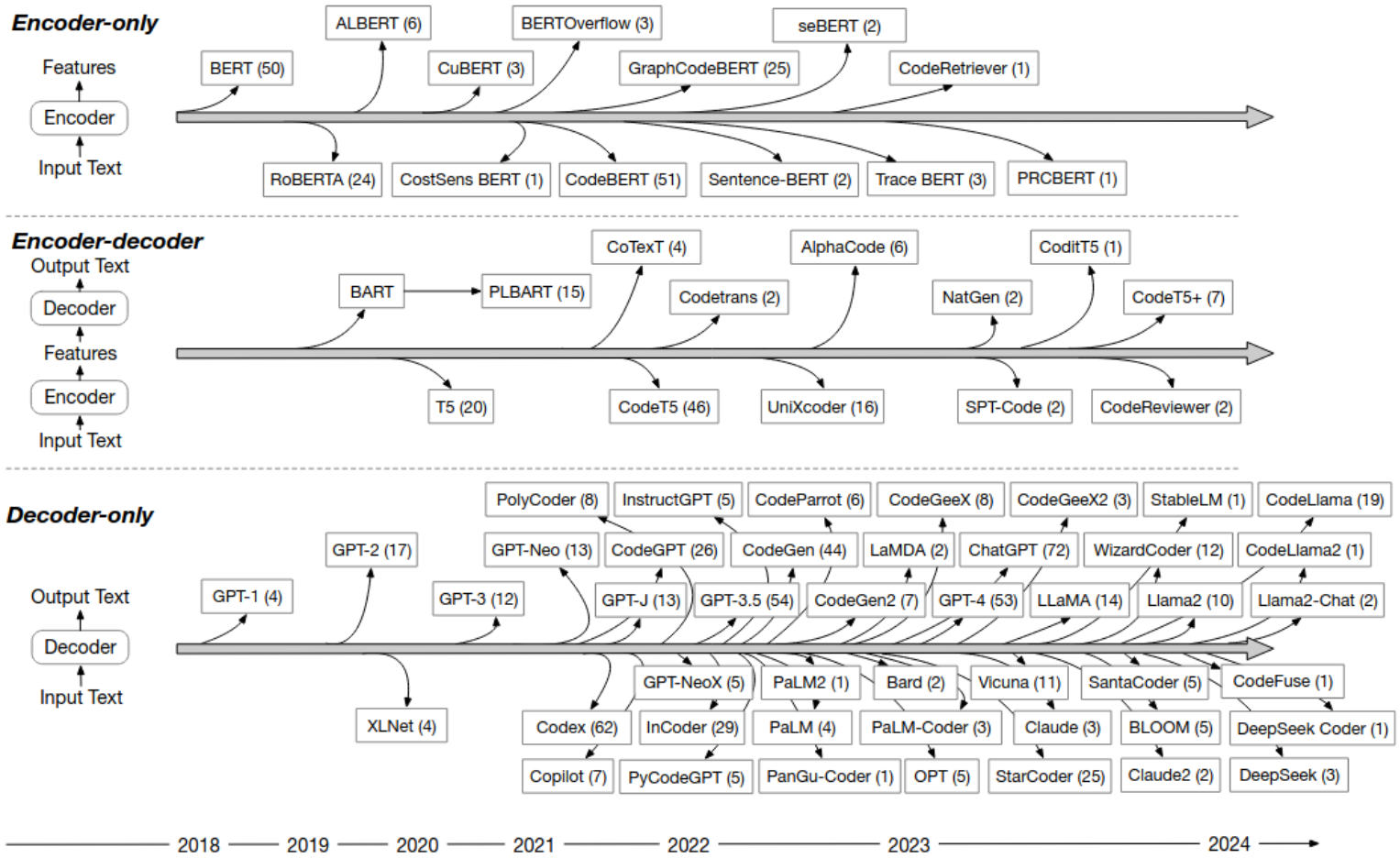


(a) Distribution of papers across venues.

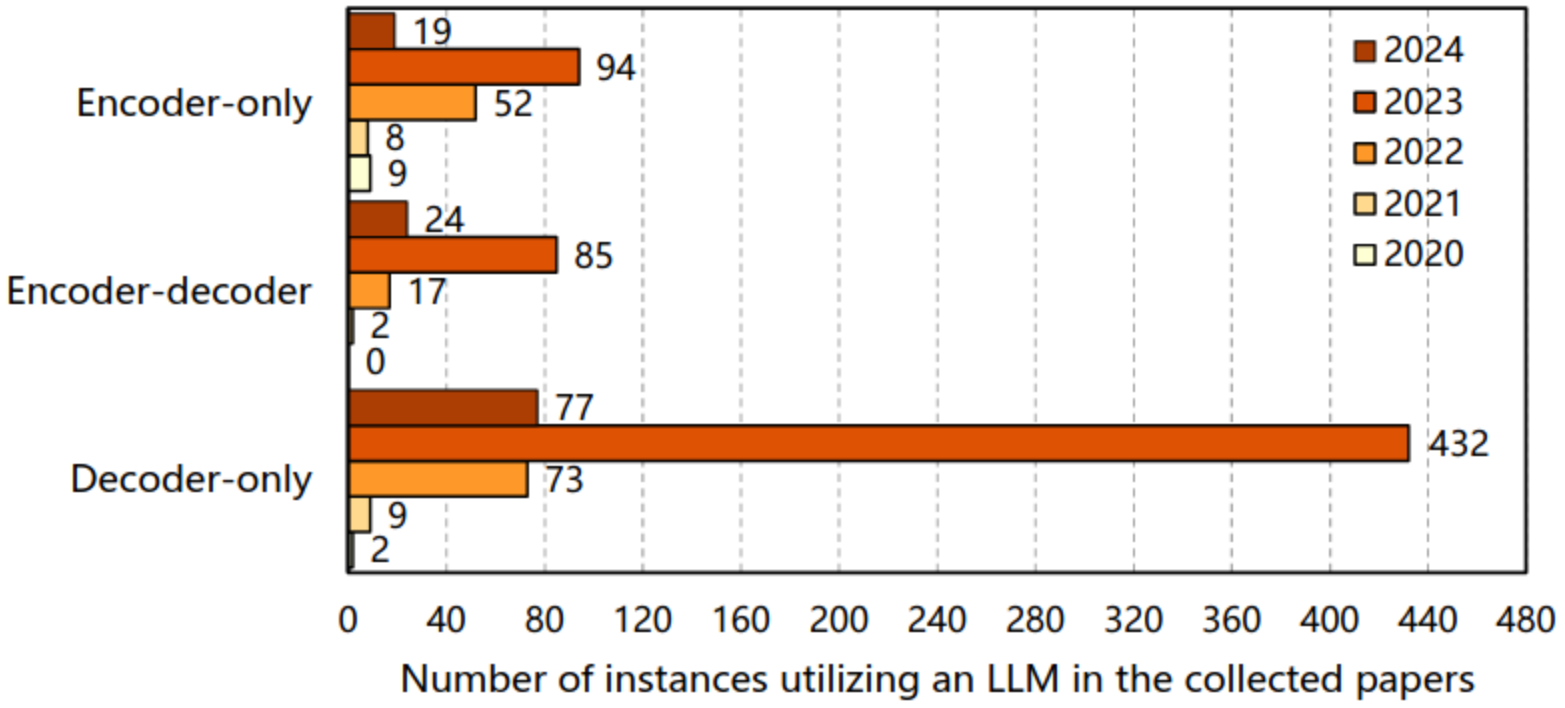


(b) Distribution of papers over years.

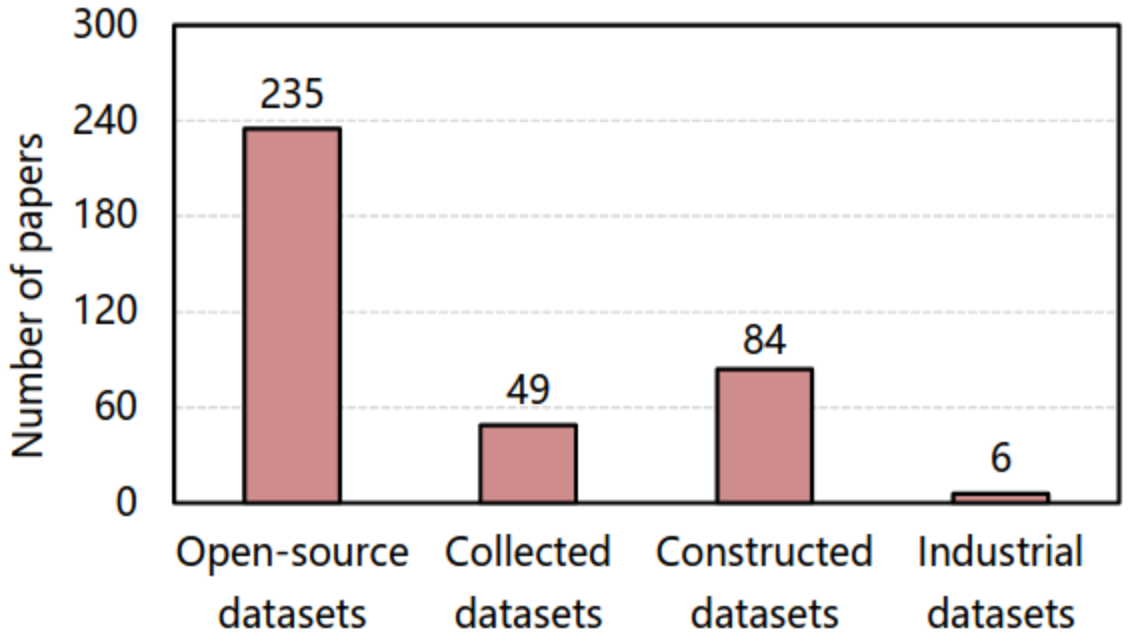
Taxonomy of the LLMs



Trends in the application of LLMs with different architectures in SE tasks over time.



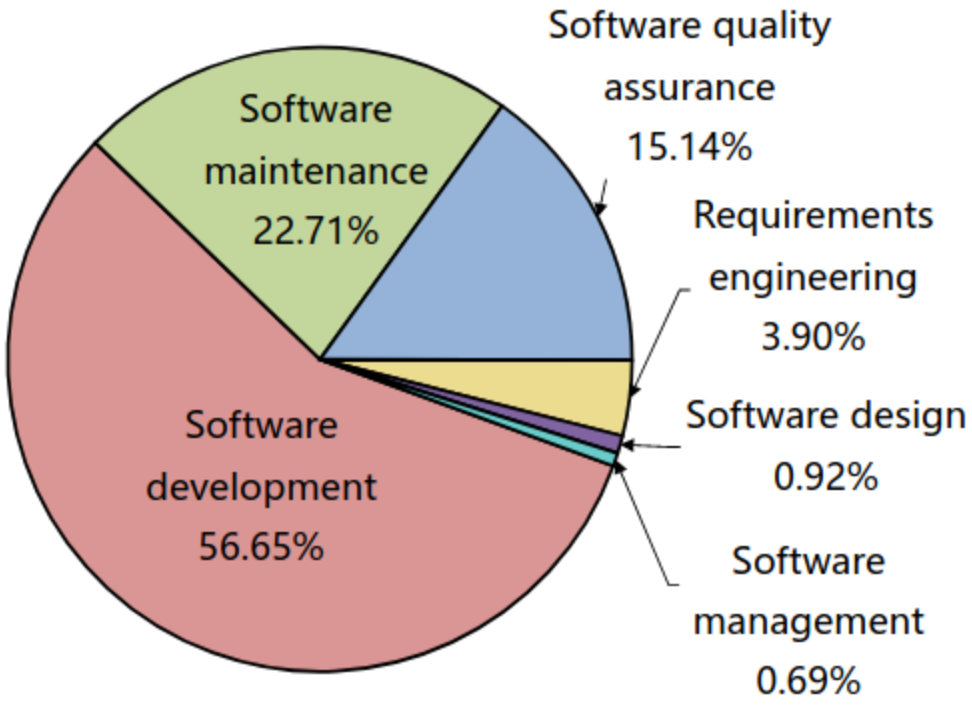
The collection strategies of LLM-related datasets.

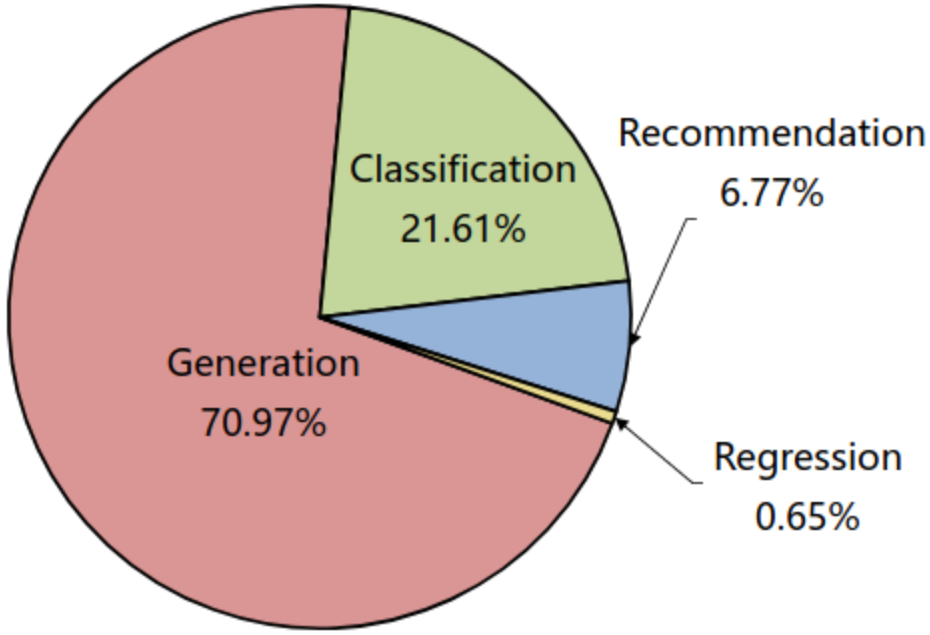


The data preprocessing procedure for code-based datasets



Distribution of LLM utilization across different SE activities and problem types





Challenges And Opportunities

- Model size and deployment
- Data dependency
- Ambiguity in code generation
- LLM Generalizability.
- LLM Evaluation
- LLM Interpretability, Trustworthiness, and Ethical Usage
- Code-specialized LLMs in SE
- Task-specific model training

Conclusion

